

The Quantum-Readiness Procurement Gap: government demand for post-quantum cryptography measured against artificial intelligence, public-key infrastructure and digital identity

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Draft v3, 2026-07-04. Dataset: Zenodo 10.5281/zenodo.21192405 (CC-BY-4.0) (Sokolov 2026). Observatory: <https://procurement.eatf.eu>

Abstract

Modern digital government rests on a stack of trust technologies: public-key infrastructure and the post-quantum cryptography that must succeed it, the digital identity that establishes who is authorised to act, and the artificial intelligence that increasingly makes the decisions in between. Governments have finalized the cryptographic standards for the post-quantum transition ([National Institute of Standards and Technology 2024b](#), [2024a](#), [2024c](#)) and have set formal goals for migrating away from quantum-vulnerable cryptography ([The White House 2022](#); [Office of Management and Budget 2022](#); [European Commission 2024](#)). Whether that commitment to reinforce the foundation of the stack is yet visible in what governments buy is an open, and measurable, question. We assemble a live, provenance-verified corpus of 4,401 public-sector procurement records across four trust-technology domains (artificial intelligence, post-quantum cryptography, public-key infrastructure, and eIDAS or digital identity), drawn from four public procurement sources spanning 106 countries, and use it to measure the demand signal for post-quantum cryptography against the demand for adjacent trust technologies. We define the quantum-readiness procurement gap as the distance between that mandate and the observed procurement share, and we operationalise it as the ratio of post-quantum to public-key-infrastructure procurement volume. Post-quantum cryptography accounts for 50 of 4,401 records, or 1.14 percent of the corpus, against 1,727 records for public-key infrastructure and 1,505 for artificial intelligence, a post-quantum-to-public-key-infrastructure record ratio of about one to thirty-five. In United States federal award data, where monetary values are available, post-quantum cryptography accounts for USD 6.3 million against USD 566.7 million for public-key infrastructure, a ratio of about one to ninety. The gap holds across every region we measure and is widest in World Bank-financed procurement, where we observe no post-quantum records. We interpret this as a measurable distance between a maturing mandate and an as-yet-faint procurement signal, distinguish the competing explanations, and argue that the procurement record is an under-used and independently verifiable instrument for tracking the readiness of the state for the post-quantum transition, complementary to the deployment-side telemetry on which readiness is usually measured ([Westerbaan et al. 2025](#); [Dubey, Varshney, et al. 2026](#)). The corpus is built so that fabrication is structurally impossible, and every figure reported here is recomputable from the published data.

1. Introduction

Governments are building digital services at speed, and everything they build stands on the same foundation: the cryptography that authenticates a machine or a document, the identity systems that decide who may act, and, increasingly, the automated systems that make decisions on the state’s behalf. That foundation has a known weakness. The public-key cryptography at its base will be broken by a sufficiently capable quantum computer, and unlike an ordinary security flaw the response has been scheduled by policy rather than triggered by an incident. Governments have been told, with deadlines, to migrate to quantum-safe cryptography. This article asks a simple question about that instruction. When we look not at what governments say, and not at what is already deployed on the public web, but at what they are actually buying, is the foundation being reinforced? The answer, on the present evidence, is that it is almost not.

A cryptographically relevant quantum computer would break the public-key cryptography that protects most digital communication and stored records today. The response of the standards community has moved from research to instruction. In August 2024 the United States National Institute of Standards and Technology finalized its first three post-quantum cryptographic standards, ML-KEM, ML-DSA and SLH-DSA, published as FIPS 203, 204 and 205 ([National Institute of Standards and Technology 2024b](#), [2024a](#), [2024c](#)). United States executive guidance has set a policy goal of mitigating quantum risk as much as feasible by 2035 ([The White House 2022](#)) and, through the Office of Management and Budget, has directed federal agencies to inventory and prioritize their cryptographic migration ([Office of Management and Budget 2022](#)), while the National Security Agency’s Commercial National Security Algorithm Suite 2.0 sets earlier adoption milestones for national security systems. The European Union has issued a parallel call for a coordinated national transition roadmap ([European Commission 2024](#)). The threat model that motivates this urgency, often described as harvest-now-decrypt-later, assumes that an adversary can record encrypted traffic today and decrypt it once a quantum computer becomes available, which makes the effective deadline earlier than the arrival of the machine, and which recent work has begun to assess for practical feasibility ([Blanco-Romero et al. 2026](#)).

Policy of this kind is meant to change behaviour, and one of the most concrete behaviours a government exhibits is what it buys. Procurement notices and contract awards are public in most jurisdictions, are timestamped, and name the buyer and the object of purchase. They are, in principle, a leading indicator of whether a mandate has begun to move money, and public procurement microdata has been used to trace innovation and policy signals in exactly this way ([Demircioglu and Vivona 2021](#); [Hickok 2024](#)). Yet the post-quantum transition has so far been measured almost entirely from the deployment side, through telemetry on the fraction of encrypted connections that negotiate a post-quantum key exchange ([Westerbaan et al. 2024, 2025](#); [Dubey, Varshney, et al. 2026](#)). The procurement side, the purchasing and budget signal that precedes deployment, is comparatively unread, in part because it is fragmented across national systems and coding schemes, and in part because assembling a cross-domain, cross-jurisdiction view of it has historically required trusting aggregators whose provenance is opaque.

We take the procurement record seriously as a measurement instrument. Stated in one sentence, the advance of this article is a first reproducible, provenance-verified measurement of the quantum-readiness procurement gap across four domains and 106 countries: post-quantum cryptography is about 1.14 percent of visible trust-technology procurement and roughly one part in ninety of tagged United States federal award value, and that smallness survives cuts by region, by monetary value, and by buyer base. The article makes three contributions. We assemble and publish a live, provenance-verified corpus of trust-technology procurement across four domains, built so that a record cannot exist without a verifiable live source. We use that corpus to measure the demand signal for post-quantum cryptography and to place it in the context of the adjacent trust technologies that governments are buying at the same time. And we report the gap itself, with its competing explanations distinguished and its measurement independently checkable.

We are careful about what this measures. The corpus records demand signals, meaning what buyers advertised or awarded, and it is a curated coverage sample rather than a census of all government procurement. The finding is descriptive, and we make no causal claim linking mandate to procurement. We claim only that the visible, nameable, purchasable demand for post-quantum cryptography is, at the time of measurement, very small relative to the mandate and relative to adjacent trust-technology demand, and that this is worth measuring and watching.

2. Background

2.1 The four domains as a digital-trust stack

We measure four domains together, and the choice is deliberate rather than a convenience sample of popular technologies. The four are the procurement-visible layers of a single digital-trust stack, the machinery through which a state constitutes and delegates trust in its digital operations. Public-key infrastructure is the incumbent cryptographic root: the certificates, key management and hardware security modules that let one system prove to another that a message, a document or a server is authentic. Post-quantum cryptography is the future of that same root: it is not a separate market but the migration that keeps the cryptographic base from collapsing once a quantum computer can break today's public-key algorithms, which is precisely why we read it against public-key infrastructure rather than in isolation. Digital identity, including the European eIDAS framework and its qualified trust services, is the layer above the root that establishes who, whether a person or an organisation, is authorised to act, and it rests on the cryptographic base beneath it. Artificial intelligence sits at the top: the systems that increasingly make or shape the decisions the state and its citizens must be able to trust and to hold to account.

Three properties make these four a coherent object of measurement. They are interlocking rather than independent: the post-quantum migration is a migration of public-key infrastructure, digital identity depends on that infrastructure, and the identity and decision layers increasingly call on each other, so a weakness in one layer is a weakness in the stack. They are each, at the same time, the subject of an explicit and recent government-level policy instrument that drives procurement: the post-quantum standards and mandates discussed below, the European eIDAS 2.0

regulation and digital-identity-wallet programme for identity, and the emerging body of artificial-intelligence regulation and public-sector strategy for the decision layer (Hickok 2024). And they are each identifiable in the procurement record through classification codes and domain vocabulary, which makes the stack observable through the same instrument. Reading post-quantum cryptography against the other three layers is therefore not an arbitrary comparison. It asks whether a foundation that the identity and decision layers are being built upon at speed is itself being reinforced at a commensurate rate.

2.2 Standards and mandates

The post-quantum transition is unusual among cryptographic migrations in that its schedule is set by policy rather than by the discovery of a break in a deployed algorithm. The three finalized National Institute of Standards and Technology standards give implementers concrete targets (National Institute of Standards and Technology 2024b, 2024a, 2024c). United States national-security guidance and the Office of Management and Budget convert those targets into goals, deadlines and inventory obligations for federal and national-security systems (The White House 2022; Office of Management and Budget 2022), and a joint agency information sheet has set out the practical migration steps expected of critical-infrastructure owners (Cybersecurity and Infrastructure Security Agency and National Security Agency and National Institute of Standards and Technology 2023). Comparable coordination is under way in the European Union (European Commission 2024). The practical consequence is that, for a large class of public buyers, quantum-safe cryptography moved from optional to expected over the period our corpus covers.

2.3 Procurement as a signal, and why it is hard to read

Public procurement is one of the most complete administrative records a state produces, and in the European Union above-threshold tenders are published centrally and in structured form. Researchers have used procurement microdata to study innovation adoption and policy effects (Demircioglu and Vivona 2021), to build open research databases of award notices (Potin et al. 2023; Omari et al. 2025), and to examine how governments acquire and govern emerging technologies such as artificial intelligence (Hickok 2024; Johnson et al. 2024). The signal is nonetheless hard to read across domains and borders for three reasons. Coding systems differ, so a purchase of cryptographic services carries a different classification code in the European Union than a comparable purchase in the United States. Trust-technology purchases hide inside broad information-technology categories rather than carrying a dedicated code, so a keyword layer is required to separate an artificial-intelligence contract from a generic software contract. And the completeness of the underlying feeds varies, so a naive cross-country count reflects the depth of each feed as much as the behaviour of each government. We address the first two problems with a full-text and code-based selection strategy, aligning records to the Open Contracting Data Standard (Open Contracting Partnership 2021), and we address the third by reporting coverage caveats alongside every cross-region comparison rather than presenting raw counts as a census.

2.4 Trustworthy measurement

The corpus that supports this article is the successor to an earlier project whose collection pipeline produced fabricated source records. That experience shaped the present design. Rather than ask readers to trust that the data was collected honestly, we built the collection so that a fabricated record cannot enter it, a property we describe in Section 3.2 and treat as a data-quality guarantee rather than a narrative. For a finding that rests on an absence, this matters: an absence is informative only if the reader can rule out that it is an artifact of a broken or dishonest pipeline.

2.5 Related work and the gap this fills

The readiness of the world for the post-quantum transition is, at present, measured almost entirely on the deployment side. Continuous telemetry on the share of encrypted web connections negotiating a post-quantum key exchange shows that adoption on the public Internet has moved from negligible to a substantial minority of connections over 2024 and 2025 ([Westerbaan et al. 2024, 2025](#)), and independent measurement studies corroborate this trajectory across servers and clients ([Dubey, Varshney, et al. 2026](#)). This deployment lens is powerful but partial: it observes cryptography that is already running, it is weighted toward the large content and platform providers that dominate connection volume, and it is largely silent on the public sector, whose systems are often not exposed as public web endpoints and whose migration is procured rather than pushed. The procurement lens we develop here is the complement. It observes a purchasing and budget signal that precedes deployment, it is weighted toward public buyers by construction, and it can be read for cryptography that has been ordered but not yet fielded. To our knowledge no prior work measures post-quantum readiness through the procurement record, nor places it against the demand for the adjacent trust technologies. Relative to prior procurement-microdata research, which is typically single-source and single-domain ([Potin et al. 2023](#); [Omari et al. 2025](#); [Johnson et al. 2024](#)), this corpus is live, provenance-verified, cross-domain and cross-jurisdiction, and the anti-fabrication construction is, we believe, novel for a procurement dataset.

3. Data and methods

3.1 Sources and scope

The corpus draws on four public procurement sources. From the European Union we harvest Tenders Electronic Daily, the central publication venue for above-threshold notices across the member states, through its version 3 application programming interface. From the United States we harvest federal award data from USAspending.gov. From the United Kingdom we harvest Contracts Finder and the Find a Tender Service. For the wider world we harvest World Bank procurement notices, which cover procurement financed by the Bank across a large set of borrower countries. Each source is public and carries an open or reuse-permitting licence, recorded per record and preserved in the aggregate release under a Creative Commons Attribution licence.

We tag each record to one or more of four domains: artificial intelligence, post-

quantum cryptography, public-key infrastructure, and eIDAS or digital identity. For domains where a coding system helps, we use only classification codes that we verified return real notices. For domains that are keyword-dominant, including post-quantum cryptography, we query source full-text indices directly for domain terms, which finds relevant notices regardless of the language of the notice. The post-quantum term set includes *post-quantum*, *post quantum*, *quantum-safe*, *quantum resistant*, *ML-KEM*, and *FIPS 203*, among others; the full selection strategy for all four domains is published with the corpus.

3.2 Anti-fabrication by construction

A record enters the corpus only if a deterministic harvest script received a live successful response from a listed source during the harvest run. There is no manual insertion path, and no language model participates in the harvest loop. Every record carries four provenance fields: the source endpoint, the fetch timestamp, the observed response status, and a SHA-256 hash of the raw upstream payload from which the record was derived. A validator rejects, at write time, any record that lacks a provenance field, that carries a future date, that names a placeholder host, or that carries a classification code outside the verified set. Records are deduplicated across harvest runs by their stable identifier. A single analysis script computes every statistic reported in this article from the committed records, so that any reported figure can be recomputed and no figure can be asserted without the records behind it. The full pipeline is open and free of third-party dependencies. Because a record cannot exist without a verifiable live source, the small post-quantum count is not the residue of a filter that silently discarded post-quantum records; it is what the four listed sources returned.

3.3 What the corpus measures, and what it does not

The corpus measures the visible demand signal, meaning notices and awards that a buyer published or made, tagged to a trust-technology domain. It is a curated coverage sample. The European Union source is the deepest and is close to complete for above-threshold tenders, which is why European member states dominate the country ranking. The United States and United Kingdom sources are keyword-sparse in the current build and undercount relative to the European Union. The World Bank source reflects development-finance procurement rather than general government procurement. Monetary values are available only for the United States award source. These properties bound our claims: we report within-source and within-region shares, we compare like with like, and we treat every cross-region comparison as a comparison of visible procurement rather than of total national spend.

4. Results

4.1 The four-domain demand picture

The corpus holds 4,401 unique records spanning 106 countries. Table 1 gives the distribution across domains and regions. Public-key infrastructure is the largest domain with 1,727 records, ahead of artificial intelligence with 1,505 and eIDAS or digital

identity with 1,119. Post-quantum cryptography is the smallest by a wide margin, with 50 records. Grouping the three cryptography-and-identity domains together, security and identity procurement accounts for 2,896 records against 1,505 for artificial intelligence, so the trust-infrastructure domains together outweigh artificial intelligence in the visible record by close to two to one.

Table 1. Records by region and domain. Regional totals are shaped by source depth (Section 3.3). A dagger (†) marks a coverage zero, where the region is thinly covered, as distinct from a value that is simply small; the United States eIDAS zero is a genuine zero within an otherwise well-populated source, whereas the United Kingdom and World Bank post-quantum zeros are coverage zeros.

Region	AI	PKI	PQC	eIDAS	Total
EU	1,200	1,094	33	963	3,290
US	73	471	17	0	561
Global (World Bank)	230	162	0†	155	547
UK	2	0†	0†	1	3
Total	1,505	1,727	50	1,119	4,401

The regional totals reflect the near-completeness of the Tenders Electronic Daily feed rather than a claim that European governments buy the most trust technology in absolute terms. The comparison we draw below is within region wherever possible, which neutralizes that source-depth effect.

4.2 The quantum-readiness procurement gap

Figure 1 shows the gap in the three independent cuts on which our claim rests. Post-quantum cryptography accounts for 50 of 4,401 records, or 1.14 percent of the corpus (panel A), a post-quantum-to-public-key-infrastructure record ratio of about one to thirty-five. The share is small in every region we measure and does not depend on the source-depth caveat, because it is a within-region ratio (panel B): 17 of 561 records in the United States, or 3.03 percent; 33 of 3,290 records in the European Union, or 1.0 percent; and 0 of 547 records in World Bank-financed procurement. The United States share, though the highest of the three, still places post-quantum cryptography an order of magnitude below public-key infrastructure within the same source and jurisdiction.

The monetary comparison, available for the United States award source, is starker than the record count (panel C). Post-quantum cryptography accounts for USD 6.3 million of tagged federal award value, against USD 566.7 million for public-key infrastructure and USD 154.3 million for artificial intelligence. Public-key infrastructure, the incumbent cryptographic infrastructure that the post-quantum transition is meant to succeed, attracts roughly ninety times the award value of its designated successor in the same records. Table 2 states the three cuts with their explicit denominators.

Table 2. The quantum-readiness gap in three cuts.

Figure 1. The quantum-readiness procurement gap, in three cuts (counts, within-region share, and award value).

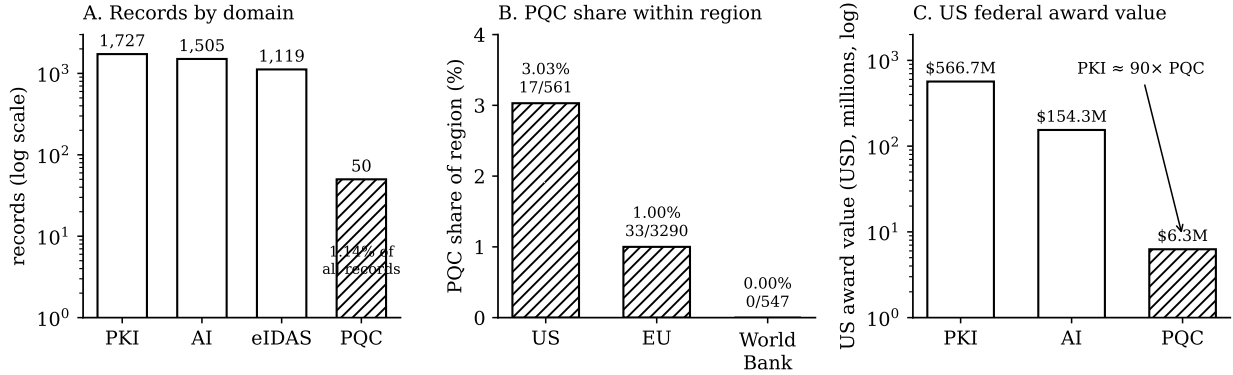


Figure 1: Figure 1. The quantum-readiness procurement gap, in three cuts. Post-quantum cryptography is the hatched outlier in each panel. (A) Records by domain, log scale. (B) Post-quantum share within each region. (C) United States federal award value, log scale, with the public-key-infrastructure to post-quantum ratio annotated. Every value is read from the published analysis file.

Cut	Post-quantum	Comparator	Ratio
Corpus record share	50 / 4,401 (1.14%)	PKI 1,727	≈ 1 : 35
Within-region share (US)	17 / 561 (3.03%)	PKI 471 (US)	≈ 1 : 28
Within-region share (EU)	33 / 3,290 (1.0%)	PKI 1,094 (EU)	≈ 1 : 33
US federal award value	USD 6.3 M	PKI USD 566.7 M	≈ 1 : 90

A finding that survives the overall share, the within-region ratio, and the monetary cut is unlikely to be an artifact of any single measurement choice. The reader can reproduce each cut directly: the corpus share is $50 / 4,401 = 1.14$ percent; the United States share is $17 / 561 = 3.03$ percent; the European Union share is $33 / 3,290 = 1.0$ percent; and the value ratio is $566.7 / 6.3 = 89.95$. Each of these is computed by the published analysis script from the committed records.

4.3 Who is buying post-quantum cryptography

The 50 post-quantum records are not evenly distributed, and the early movers are informative. In the European Union, the most frequent post-quantum buyer is Poland's Centralny Ośrodek Informatyki, the state information-technology centre, followed by the Netherlands Enterprise Agency and the Polish National Centre for Research and Development, with further records from a German research university, a Croatian public institute, and the European Cybersecurity Competence Centre. In United States data the Department of Defense is the most frequent post-quantum buyer, consistent with the fact that national-security systems carry the earliest migration milestones, and one of its awards names the harvest-now-decrypt-later threat explicitly. The pattern is one of a small set of technically sophisticated, mandate-exposed public bodies moving first, rather than a broad base of buyers. Each named buyer resolves to one or more specific records in the published dataset, so these qualitative claims meet the same verification bar as the counts.

By contrast, the most frequent buyers in the large domains are broad and operational. In public-key infrastructure they are interior ministries and defence, homeland-security and state departments. In artificial intelligence they include a German statutory health insurer, defence, an agricultural university, and the European Commission's communications-technology directorate. The contrast between a narrow, specialist post-quantum buyer base and a broad, operational buyer base for the incumbent domains is itself part of the finding.

4.4 Trajectory

The dated portion of the corpus clusters in recent years, with 376 records dated 2022, 344 in 2023, 496 in 2024, 1,191 in 2025, and 886 in the year-to-date portion of 2026; the dated records sum to 3,293, with the remainder either undated in the source or falling outside this window. We caution that yearly totals are shaped by the publication windows of the sources and by our collection window, so the rise should be read as a coverage-and-activity signal rather than as a precise growth rate. Within that caveat, the record is consistent with rising overall trust-technology procurement over the period in which the post-quantum mandate matured, which makes the flatness of the post-quantum signal against that rising baseline the more notable.

5. Discussion

5.1 Competing explanations for the gap

A near-absence in a procurement record admits several explanations, and honest measurement means naming them.

The first is a genuine lag. Mandates with milestones through the early-to-mid 2030s may simply not have moved procurement money yet, with buyers still in the inventory-and-planning phase that the executive guidance prescribes before purchasing ([Office of Management and Budget 2022](#); [Cybersecurity and Infrastructure Security Agency and National Security Agency and National Institute of Standards and Technology 2023](#)). Under this reading the gap is real and the procurement signal is an early and

truthful indicator that migration has not yet reached the buying stage at scale.

The second is procurement invisibility. Post-quantum migration may be happening inside contracts that are not labelled as post-quantum, for example as a routine feature of a broader information-security services contract, a software update, or a hardware refresh, in which case the demand exists but does not surface as a distinctly nameable procurement. Our keyword and full-text approach would miss such embedded demand. This explanation reduces the strength of the absence claim, and we flag it prominently. It does not, however, dissolve the finding, because the same embedding argument applies to public-key infrastructure and artificial intelligence, which nonetheless surface in large numbers, so the differential visibility of post-quantum cryptography relative to adjacent domains still requires explanation.

The third is that post-quantum cryptography is not yet a tracked category in some procurement systems, which is the most plausible reading of the zero we observe in World Bank-financed procurement, where a development-finance buyer may not yet classify or require quantum-safe cryptography at all. Under this reading the zero is informative about the diffusion of the mandate beyond the standard-setting jurisdictions rather than about activity within them.

These explanations are not mutually exclusive. The measurement stands regardless of which combination holds: the visible, labelled, purchasable demand for post-quantum cryptography is very small relative to both the mandate and the adjacent domains, and the procurement record makes that smallness precise and watchable.

5.2 Policy implications, by explanation and actor

Each explanation maps to a different actor and a different corrective, which is what makes the distinction worth drawing. If the gap reflects a genuine lag, the corrective sits with the bodies that own the migration timeline: an interim procurement milestone from the Office of Management and Budget or the European coordination process, requiring agencies to report quantum-safe purchasing and not only inventory status, would convert a distant deadline into a near-term buying obligation and make the readiness signal actionable well before the deadline. If the gap reflects procurement invisibility, the corrective sits with contracting authorities: adding an explicit cryptographic-requirement line item, or a post-quantum classification tag, to procurement notices would make embedded quantum-safe demand auditable and therefore governable. If the gap reflects non-diffusion beyond the standard-setting jurisdictions, the corrective sits with the multilateral development banks and standards bodies: adopting a post-quantum requirement tag in development-finance procurement would carry the mandate into the borrower countries where our corpus currently shows a zero. In every case, making the procurement signal continuously observable, which the live observatory does, gives policy-makers and oversight bodies an instrument for tracking readiness that does not depend on self-reported migration status, and that complements the deployment-side telemetry ([Westerbaan et al. 2025](#)) on which readiness has until now been measured.

6. Limitations

Coverage is a curated demand-signal sample and not a population census. The European Union source is the deepest, and the United States and United Kingdom sources undercount, so absolute cross-region comparisons are comparisons of visible procurement rather than of total spend, and our central within-region ratios are chosen precisely to avoid that pitfall. Domain tagging uses keyword and code matching and can both miss embedded demand, as Section 5.1 discusses, and, in principle, over-match on incidental mentions, though our inspection of the buyer base for each domain did not surface over-matching at a level that would change the finding. Monetary values exist only for the United States source. The corpus is a snapshot at the time of measurement and is designed to be re-harvested, so the figures here describe a dated build and will move as coverage deepens. Procurement is a leading and partial indicator of migration and not a measure of migration itself, and we make no causal claim linking mandate to procurement.

7. Conclusion

We set out to measure whether the maturing mandate for post-quantum cryptography is yet visible in what governments buy. Using a live, provenance-verified corpus of 4,401 trust-technology procurement records across four domains and 106 countries, we find that post-quantum cryptography accounts for 1.14 percent of the corpus and for roughly one part in ninety of tagged United States federal award value, and that the smallness of the signal holds across every region and value measure we examine, against a backdrop of large and rising procurement of the adjacent trust technologies. We call this the quantum-readiness procurement gap. It is a descriptive finding, it admits several explanations that further work can distinguish, and it rests on a measurement whose provenance the reader can verify. The procurement record is an under-used instrument for observing the readiness of the state for the post-quantum transition, complementary to the deployment-side telemetry on which readiness is usually measured, and on the present evidence that record shows a mandate that has not yet reached the point of purchase.

Data and code availability

The full corpus is published on Zenodo under a Creative Commons Attribution 4.0 licence, concept DOI 10.5281/zenodo.21192405, which resolves to the latest version (Sokolov 2026). The harvesting, validation, consolidation and analysis pipeline, the source ledger, and this article’s figure are open at <https://github.com/tyche-dev/trust-tech-procurement-observatory>. The live observatory, from which every figure in this article can be reproduced, is at <https://procurement.eatf.eu>.

Competing interests

The author declares no competing interests.

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